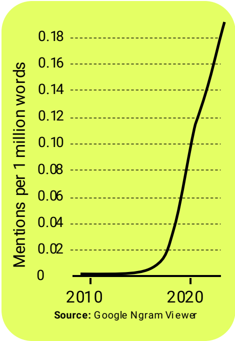
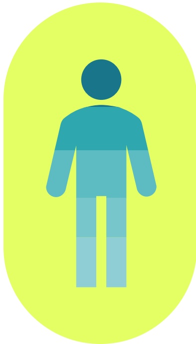
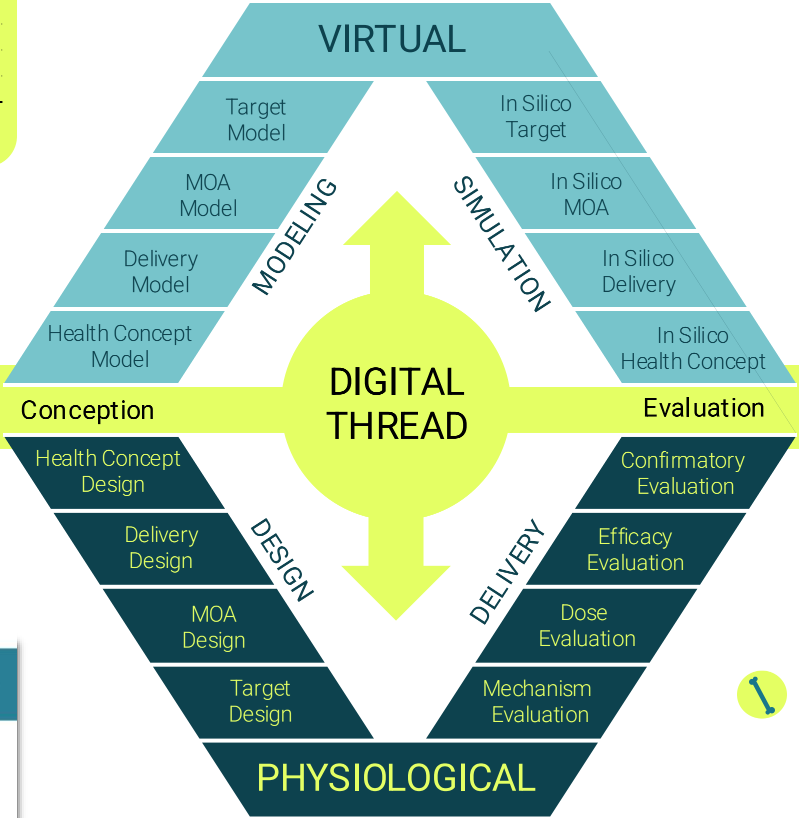
Digital Twin takes off

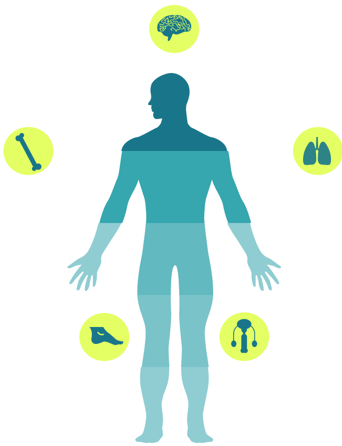


Digital Twin

Virtual patient







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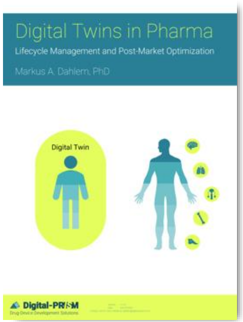


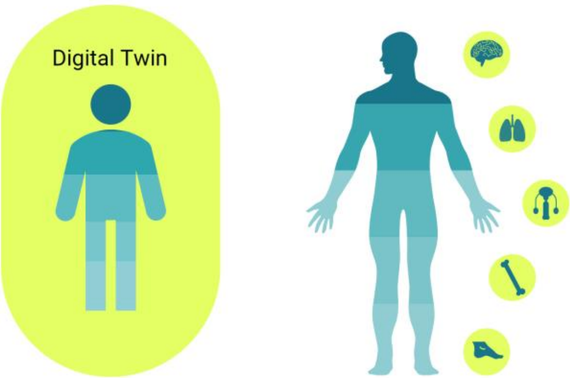
Digital-PRISM

Digital Health Development Solutions

Digital Twins in Pharma

Lifecycle Management and Post-Market Optimization

Markus A. Dahlem, PhD



Digital-PRISM
Drug-Device Development Solutions

Version: v1.0
Date: 05/05/2025
Contact me for more details at dahlem@digital-prism.com

Executive Summary

Everyone's excited about digital twins in healthcare — but not every model with real-time data is a digital twin. There is confusion what a digital twin actually is. Originally developed in engineering to optimize machines and processes, they are now positioned to transform medicine — by modeling not only physical structures, but also physiological systems, therapeutic interventions, health outcomes, and even entire patient journeys.

What defines a digital twin is not its complexity, but its capacity for control. A digital twin is dynamically coupled to its physical counterpart through continuous data exchange and evolves across its full lifecycle. This finally enables what precision medicine has long promised: care that is adaptive, feedback-driven, and truly individualized.

It is actually simple, in medicine, digital twins serve three core purposes: maintaining health, restoring it, and inventing new ways to do both. Their strength lies in replicating physiological and behavioral processes, synchronize with their human counterpart via data assimilation and and then link gained insights with intervention design and real-world validation — through what we call *twin control*: the dynamic, bidirectional coupling of model and patient across the full lifecycle of care and therapy.

To organize the models and feedback loops that constitute twin control, we adapt the *Design Diamond* from industrial manufacturing to drug–device development: a lifecycle V-model and its mirrored virtual counterpart, connected by a digital thread. This structure doesn't just align with the pharmaceutical pipeline — it upgrades it.

Digital twins transform therapeutic products into learning systems. While early R&D and clinical trial applications remain limited by regulatory inertia, their first major impact is likely post-market — where real-world data can be assimilated to refine dosing, personalize care, and deliver differentiation, even in saturated therapeutic classes. From novel therapies to generics, digital twins enable continuous optimization.

Their promise is clear: intelligent therapeutics.